Sample exam

Please note: This is just a suggested question paper and marks distribution. The actual question paper may differ in difficulty level and marks distribution

Maximum Marks: -- | Suggested Time: --

Name: _____ Roll No.: _____ Date: _____

Instructions

- Answer all questions.
- Write concise, scientifically clear answers. Label all diagrams and illustrations.
- For multiple-choice questions, select one best answer.
- Marks for each question are shown in brackets.

Section A — Objective Questions

Each question carries 1 mark

1. Which experiment produced organic monomers by exposing a simulated early-Earth gas mixture to an electric discharge? (A) Hershey–Chase (B) Miller–Urey (C) Meselson–Stahl (D) Anfinsen

2. In natural selection, which individuals are more likely to leave more offspring? (A) Always the largest (B) Those with heritable traits advantageous in that environment (C) Those that try hardest to adapt (D) All individuals equally

3. Which base is present in RNA instead of thymine? (A) Cytosine (B) Guanine (C) Uracil (D) Adenine

4. An organism with genotype Pp is described as: (A) Homozygous dominant (B) Homozygous recessive (C) Heterozygous (D) Haploid

5. Which enzyme seals the gap between DNA fragments after the RNA primer is replaced?

(a) Helicase (b) Primase (c) DNA ligase (d) Topoisomerase

Section B — Reasoning and Application

Each question carries 3 mark

6. Explain how a change in environmental conditions could alter the frequency of light- and dark-coloured moths in a population. Use the idea of natural selection in your answer. **[3]**

7. A mutation replaces a hydrophobic amino acid buried in a protein's core with a charged amino acid. Predict one possible effect on protein structure or function and explain your reasoning. **[3]**

8. Distinguish genotype from phenotype. Then define homozygous and heterozygous, using suitable allele examples. **[3]**

9. In pea plants, purple flowers (P) are dominant to white flowers (p). Cross two heterozygous purple-flowered plants (Pp × Pp). Draw a Punnett square, list the possible offspring genotypes, and state the expected genotypic and phenotypic ratios. **[3]**